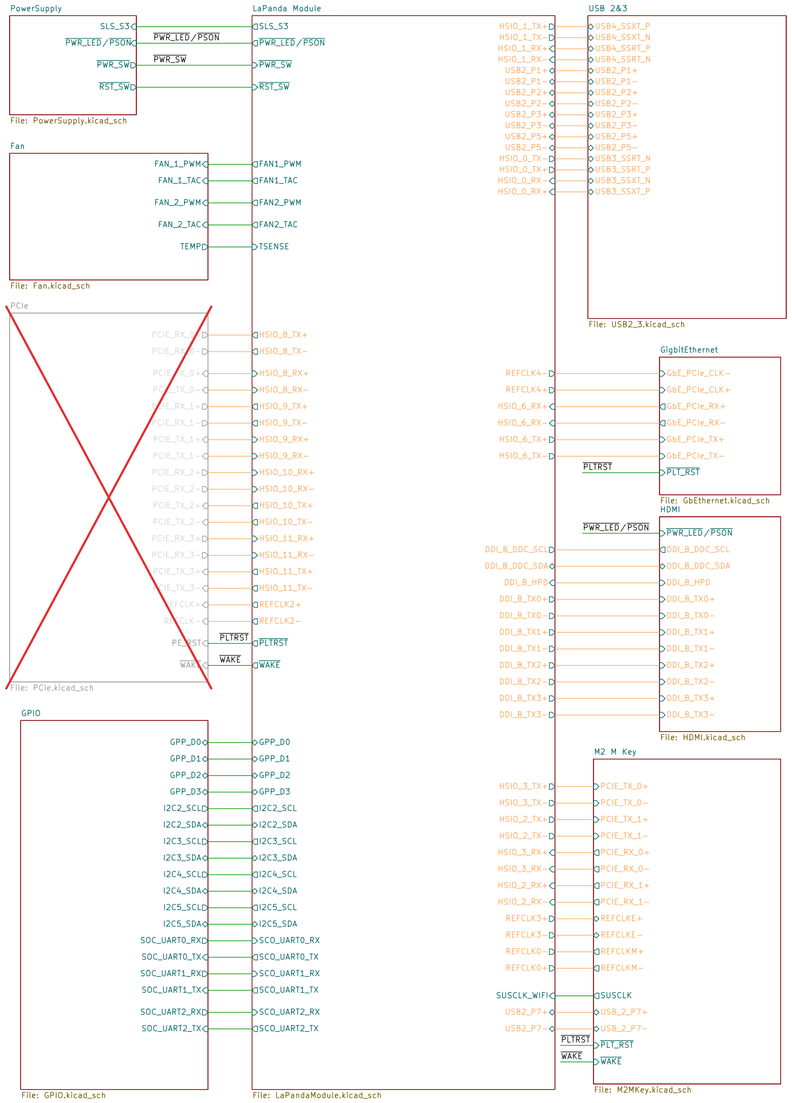


Sheet: /
File: LattepndaCourse.kicad_sch

Title: practice design

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Sleeping State Signal
-SLS_S0:
  -S0 (Working): HIGH
  -S3 (Sleeping): LOW
  -S4 (Hibernation): LOW
  -S3 (Soft Off): LOW

-SLS_S0:
  -S0 (Working): HIGH
  -S3 (Sleeping): LOW
  -S4 (Hibernation): LOW
  -S3 (Soft Off): LOW

HSIO functions are defined by the BIOS source code connot be changed in BIOS settings

Before starting design, you should choose one of the available BIOS configuration on:
https://github.com/LattePandaTeam/LattePanda-Mu
or contact technical support for custom your BIOS

The default funtions ared defined as follows:

-HSIO0: USB3.2 10Gbps
-HSIO1: USB3.2 10Gbps
-HSIO2: PCIe 3.0 x1
-HSIO3: PCIe 3.0 x1

-HSIO8: PCIe 3.0 x4 (lane 0)
-HSIO9: PCIe 3.0 x4 (lane 1)
-HSIO10: PCIe 3.0 x4 (lane 2)
-HSIO11: PCIe 3.0 x4 (lane 3)

-HSIO6: PCIe 3.0 x1

HSIO_0/HSIO_1 used for USB3.0

HSIO_2 used for M.2 M KEY

HSIO_3 used for M.2 E KEY

HSIO_8/9/10/11 used for PCIe x4

HSIO_6 used for GbE(PCIe x1)

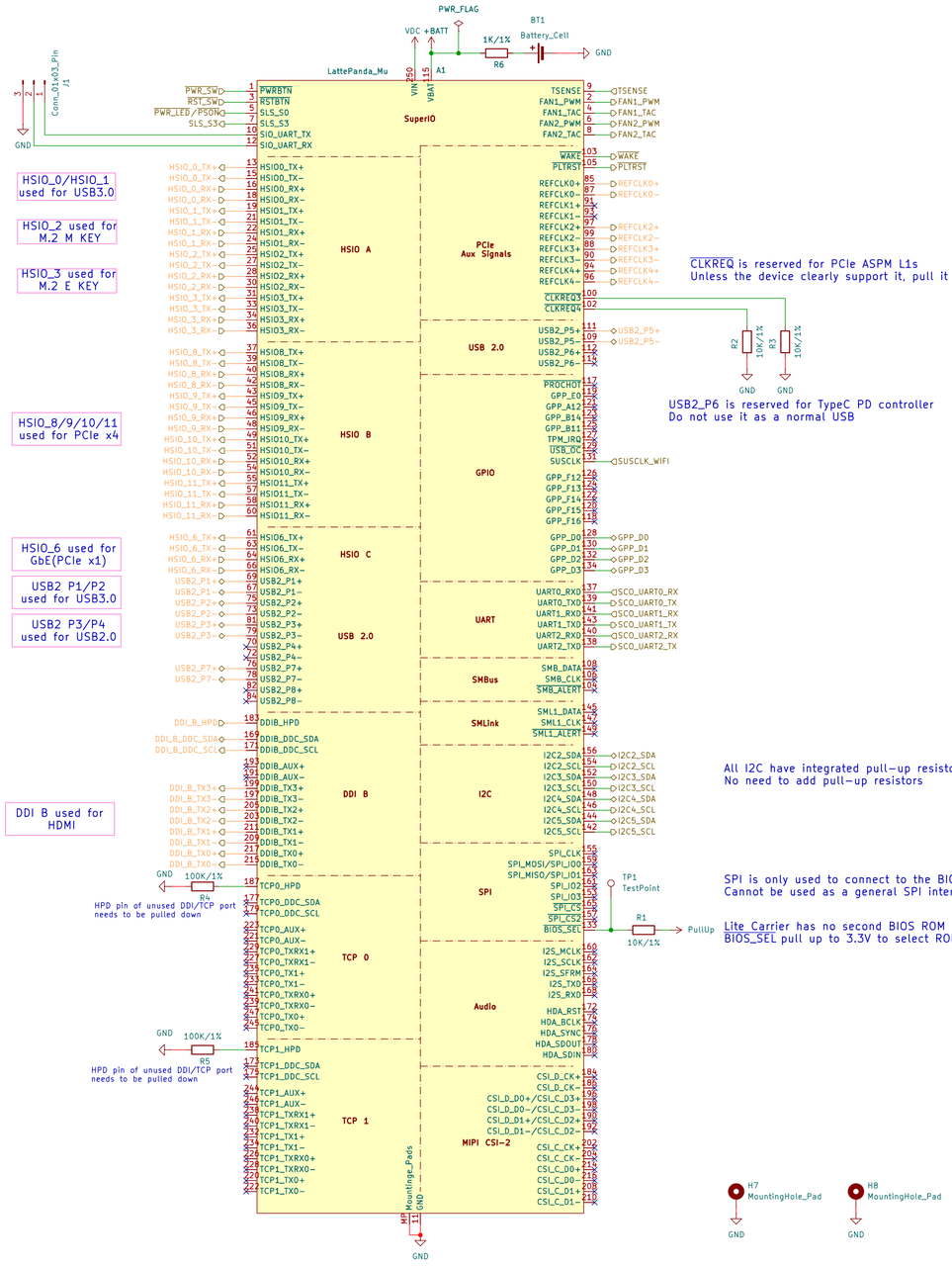
USB2 P1/P2 used for USB3.0

USB2 P3/P4 used for USB2.0

Before starting design, you should choose one of the available BIOS configuration on:
https://github.com/LattePandaTeam/LattePanda-Mu
or contact technical support for custom your BIOS

The defualt funtions are defined as follows:
-DDIA: eDP 1.4b (the 40 pin connector on Mu)
-DDIB: HDMI 2.0
-TCP0: HDMI 2.0
-TCP1: USB TypeC (needed external PD controller)

DDI B used for HDMI



CLKREQ is reserved for PCIe ASPM L1s
Unless the device clearly support it, pull it down to enhance compatiblity

USB2_P6 is reserved for TypeC PD controller
Do not use it as a normal USB

All I2C have integrated pull-up resistors.
No need to add pull-up resistors

SPI is only used to connect to the BIOS and discrete TPM.
Cannot be used as a general SPI interface.

Lite Carrier has no second BIOS ROM
BIOS_SEL pull up to 3.3V to select ROM on LattePanda Mu

Sheet: /LaPanda Module/	
File: LaPandaModule.kicad_sch	
Title: practice design	
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Power for PCIe x4 OVP813.86V

Q1
AON7262E
+12V
TP5
TestPoint
PWR_FLAG
R17
21.5K/1%
PCIe power enable LED
D12
LED
GND
C2
1uF/16V
R11
100K/1%
For soft start
R13
267K/1%
R12
267K/1%
R14
267K/1%
21.5K/1%
R15
Q2
2N7002H
U2
TL431D
R16
267K/1%
R18
21.5K/1%
Q3
2N7002H
C3
1uF/16V
GND

12V DC Input(12V 6A–10A)

J3
Barrel_Jack_Switch
PWR_FLAG
+12V
GND
TP2
TestPoint
TP3
TestPoint
D6
D7
D8
D9
D10
D11
D_Schottky
TP4
TestPoint

VDC
PWR_FLAG
PWR_FLAG
PullUp
TP8
TestPoint
R23
2.2K/1%
D13
3V3 Izt=5mA
GND

Pullup power for bios_sel

.SY8253ADC 23V 3A 500kHz

U3
LM2734X
VIN
BOOST
SW
EN
GND
FB
PWR_EN
100nF/50V
C9
L1 Vout = 0.6*(75 + 10)/10 = 5.1V
2.2uH/4.3A
R19 75K/1%
R20 10K/1%
SNP 500
C10
22uF/10V
C12
22uF/10V
C13
100nF/10V
C14
22uF/10V
C15
22uF/10V
C16
PWR_FLAG
+5V
TP6
TestPoint
GND

U4
LM2734X
PWR_EN
100nF/50V
C8
L2
2.2uH/4.3A
Vout = 0.6*(21.5 + 4.7)/4.7 = 3.446V
R21 21.5K/1%
R22 4.7K/1%
SNP 500
C17
22uF/10V
C18
22uF/10V
C19
100nF/10V
C20
22uF/10V
C21
22uF/10V
PWR_FLAG
+3.3V
TP7
TestPoint
GND

C4
22uF/10V
C5
22uF/10V
C6
22uF/10V
C7
22uF/10V
GND

JP1
Jumper_3_Bridged12
SSLS_S3
R24
100K/1%
R16
100K/1%
PWR_LED/PSON
+3.3V
R27 22K/1%
D14
LED
R25
100K/1%
Q4
2N7002
R26 10K/1%
GND

15V PD Input

J2
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
GND
SH
A1
PWR_FLAG
+15V
PWR_C_D-
PWR_C_D+
CC1
CC2
R8 10K/1%
R7 1K/1%
1uF/16V
C1
U1
CH224K
VBUS
VDD
CC1
CC2
DM
DP
CFG1
CFG2
CFG3
PG
GND
R9 56K/1%
GND
D1
D2
D3
D4
D5
ESD9B5.0ST5G
ESD9B5.0ST5G
D_Zener
D_Zener
D_Zener
GND

CH224K get DC 15V power from USB-PD adapter

Resistance on CFG1	Request-voltage
6.8KΩ	9V
24KΩ	12V
56KΩ	15V
NC	20V

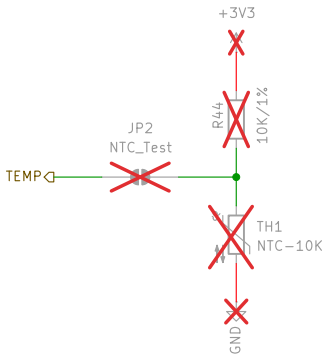
+3V3
R29 100K/1%
PWR_SW
R28 100R/1%
1uF/16V
C22
SW1
SW_Push
ESD9B5.0ST5G
GND

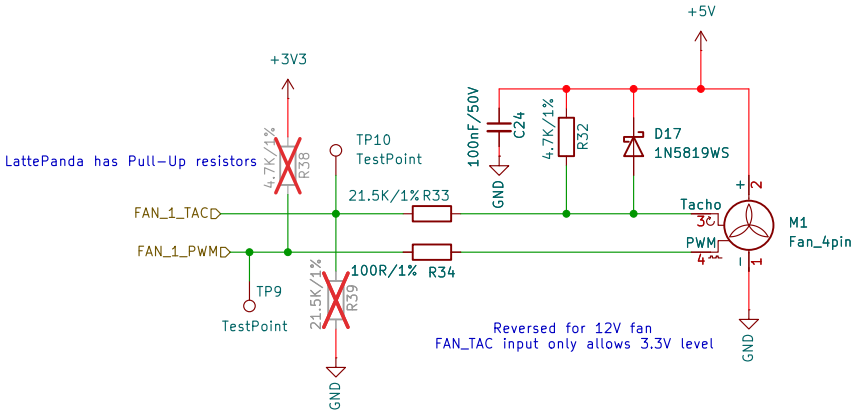
+3V3
R31 100K/1%
RST_SW
R30 100R/1%
LattePanda has Pull-Up resistors
1uF/16V
C23
SW2
SW_Push
ESD9B5.0ST5G
GND

Sheet: /PowerSupply/
File: PowerSupply.kicad_sch

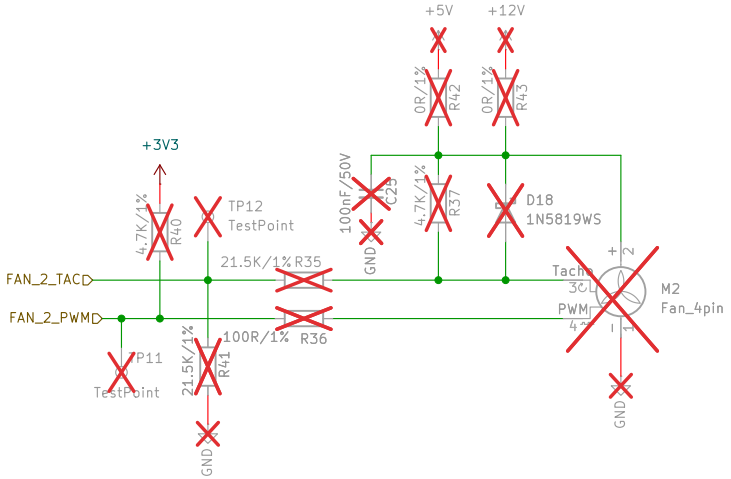
Title: practice design

Size: A2 | Date: 2026-09-30 | **Rev: 1**
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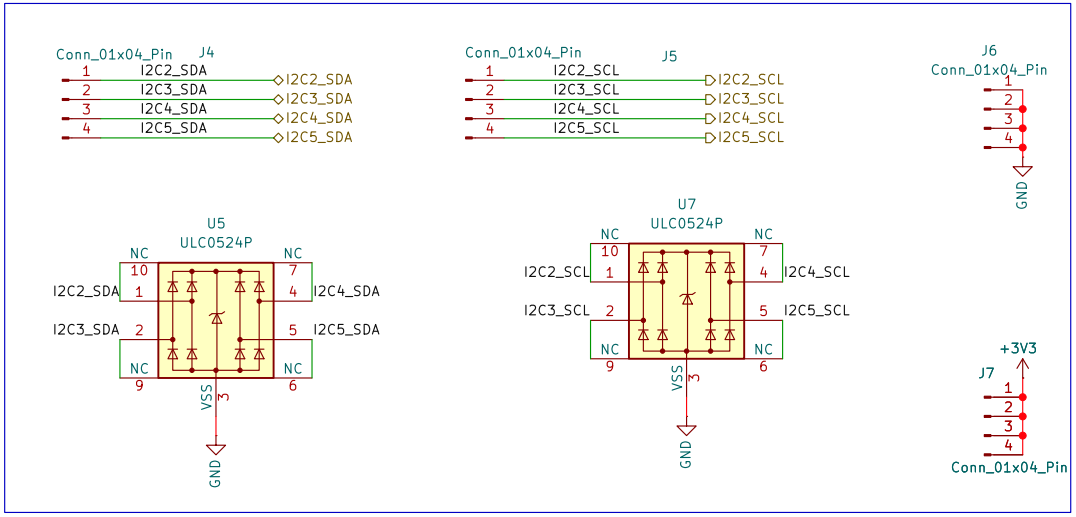
CPU FAN



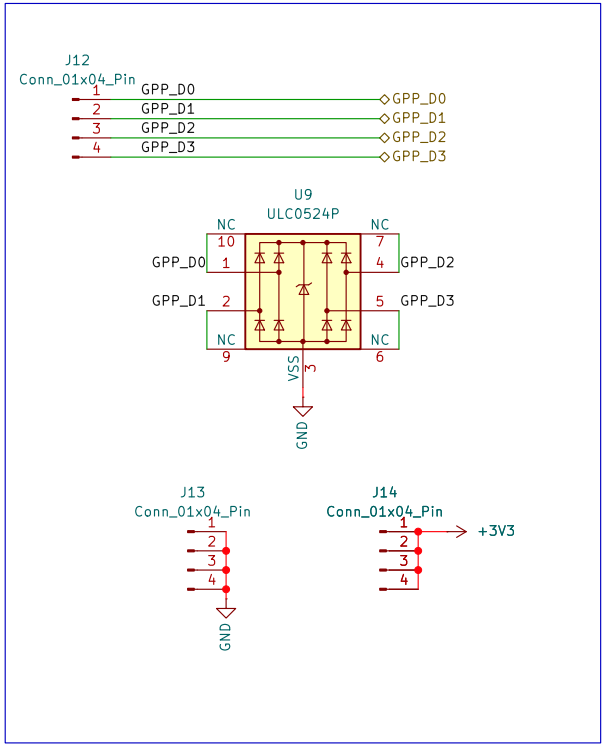
SYS FAN

Sheet: /Fan/		
File: Fan.kicad_sch		
Title: practice design		
Size: A4	Date: 2026-09-30	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 4/10

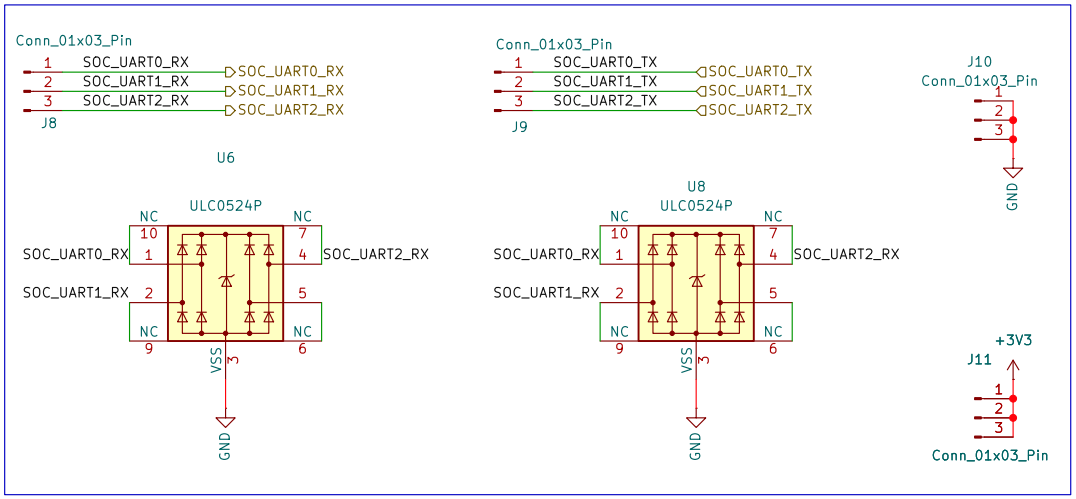
I2C



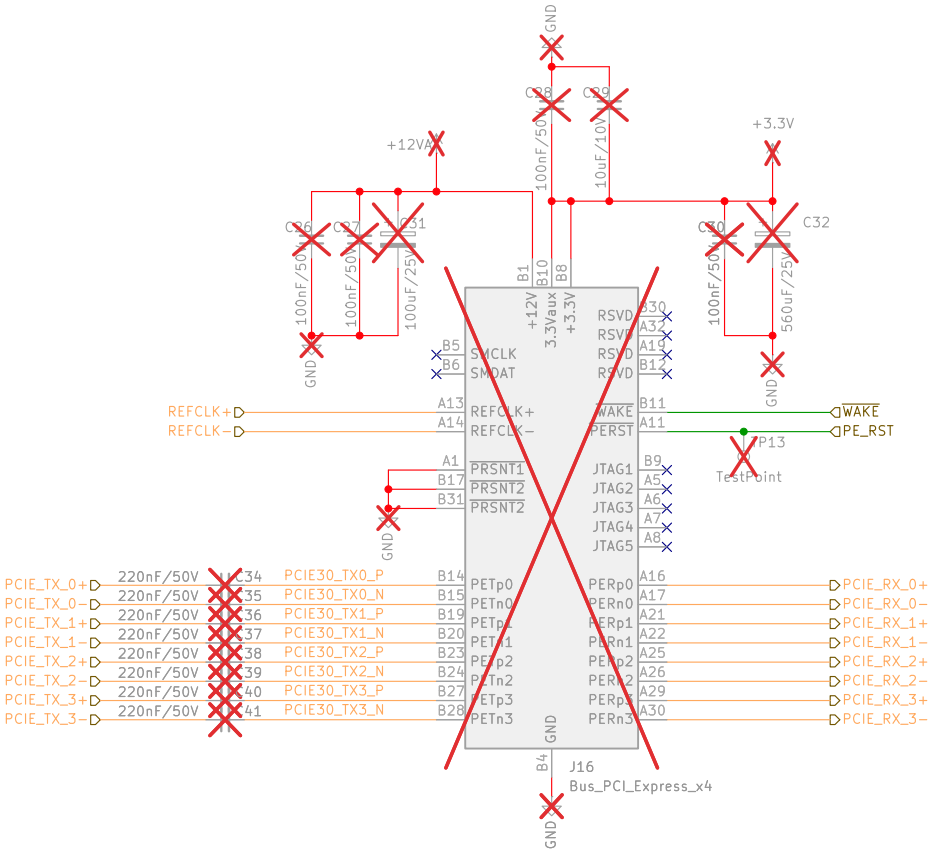
GPIO



UART



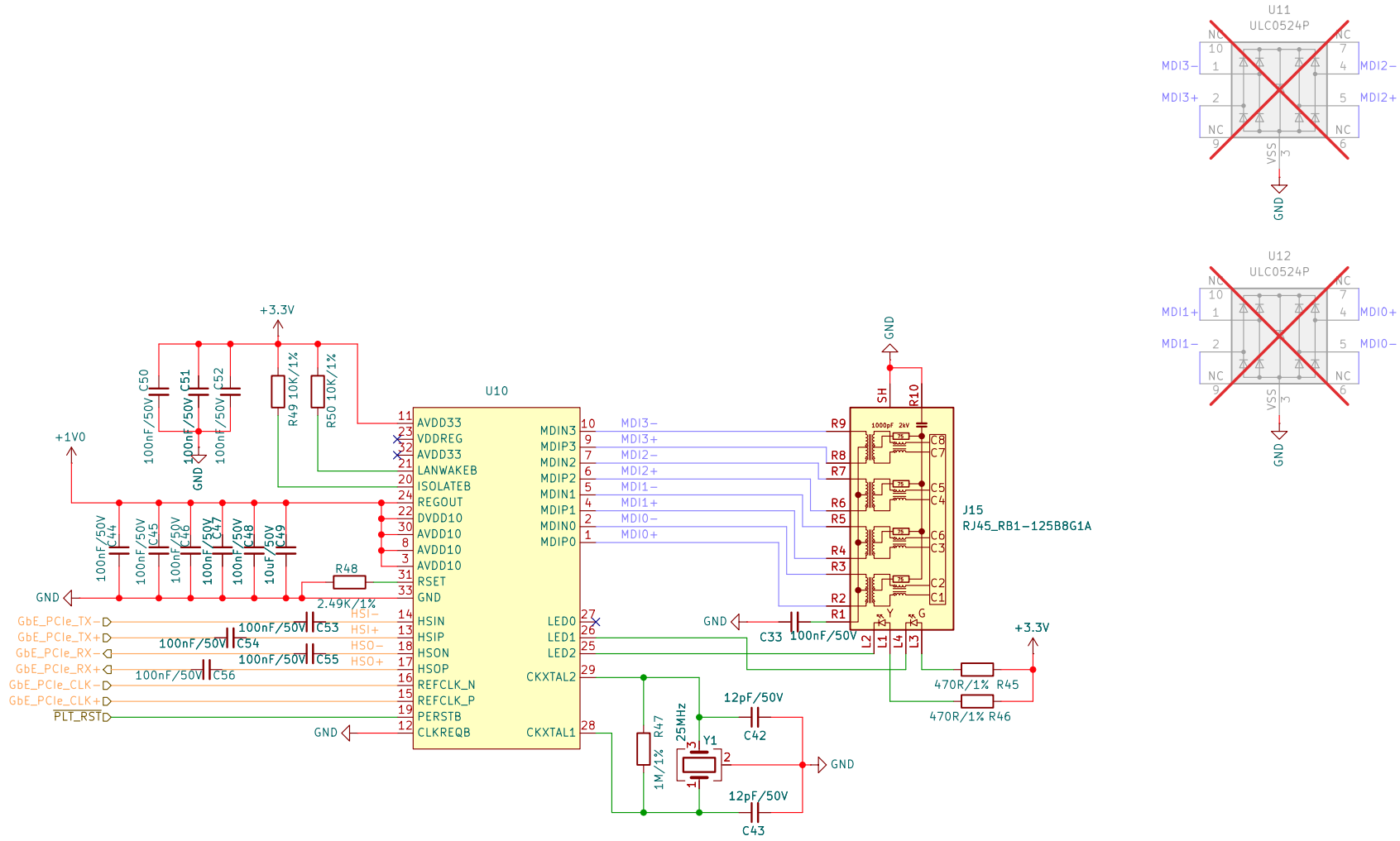
Sheet: /GPIO/		
File: GPIO.kicad_sch		
Title: practice design		
Size: A4	Date: 2026-09-30	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 5/10



Sheet: /PCIe/
File: PCIe.kicad_sch

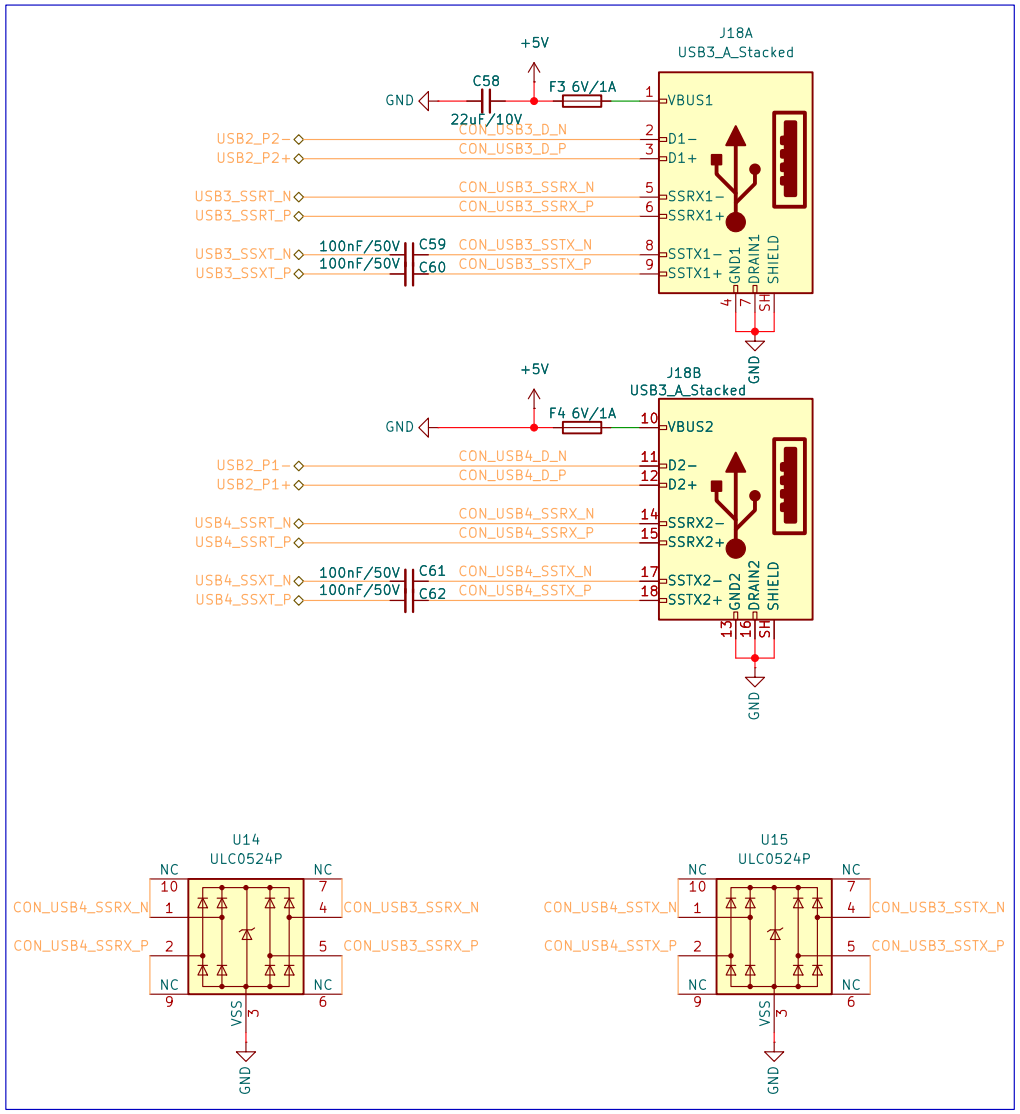
Title: practice design

Size: A4	Date: 2026-09-30	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 6/10

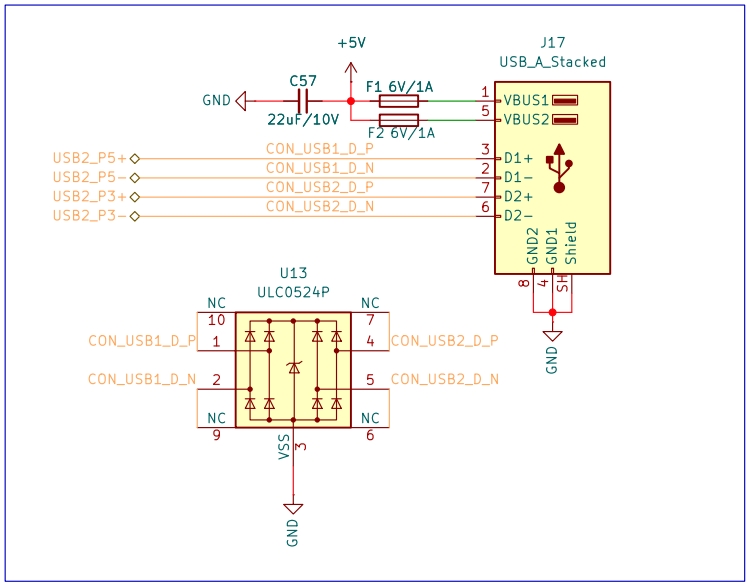


Sheet: /GigbitEthernet/		
File: GbEthernet.kicad_sch		
Title: practice design		
Size: A4	Date: 2026-09-30	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 7/10

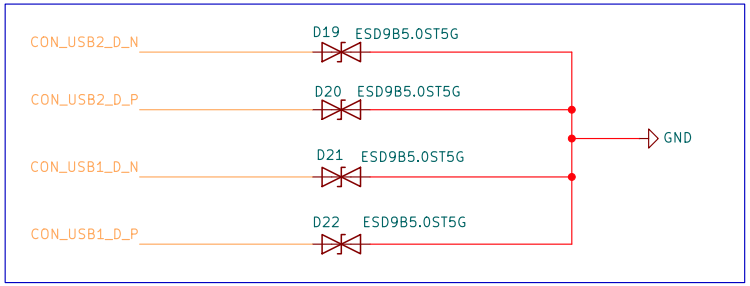
USB 3.0



USB 2.0



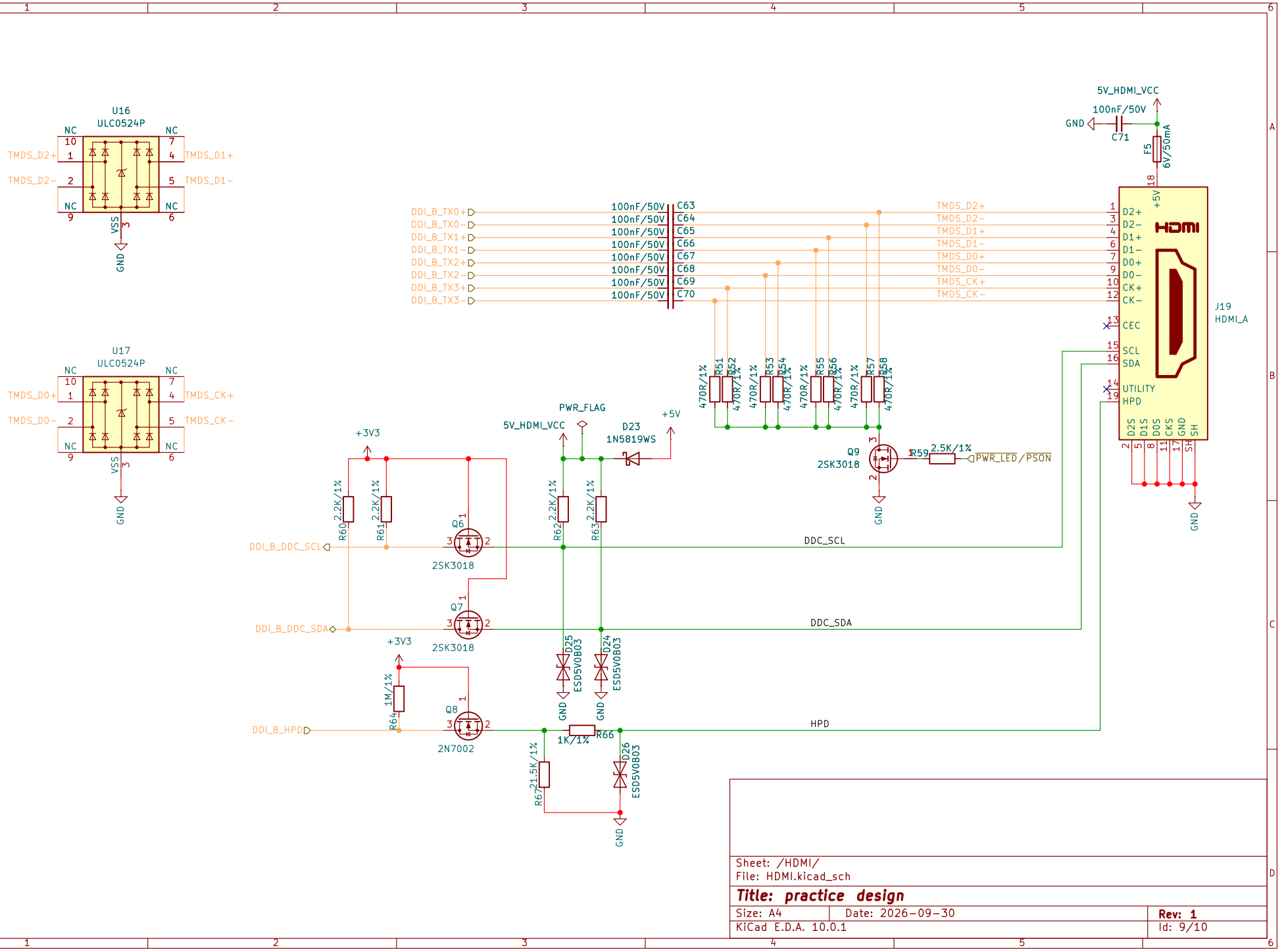
ESD Protection diods for USB 2.0 data line

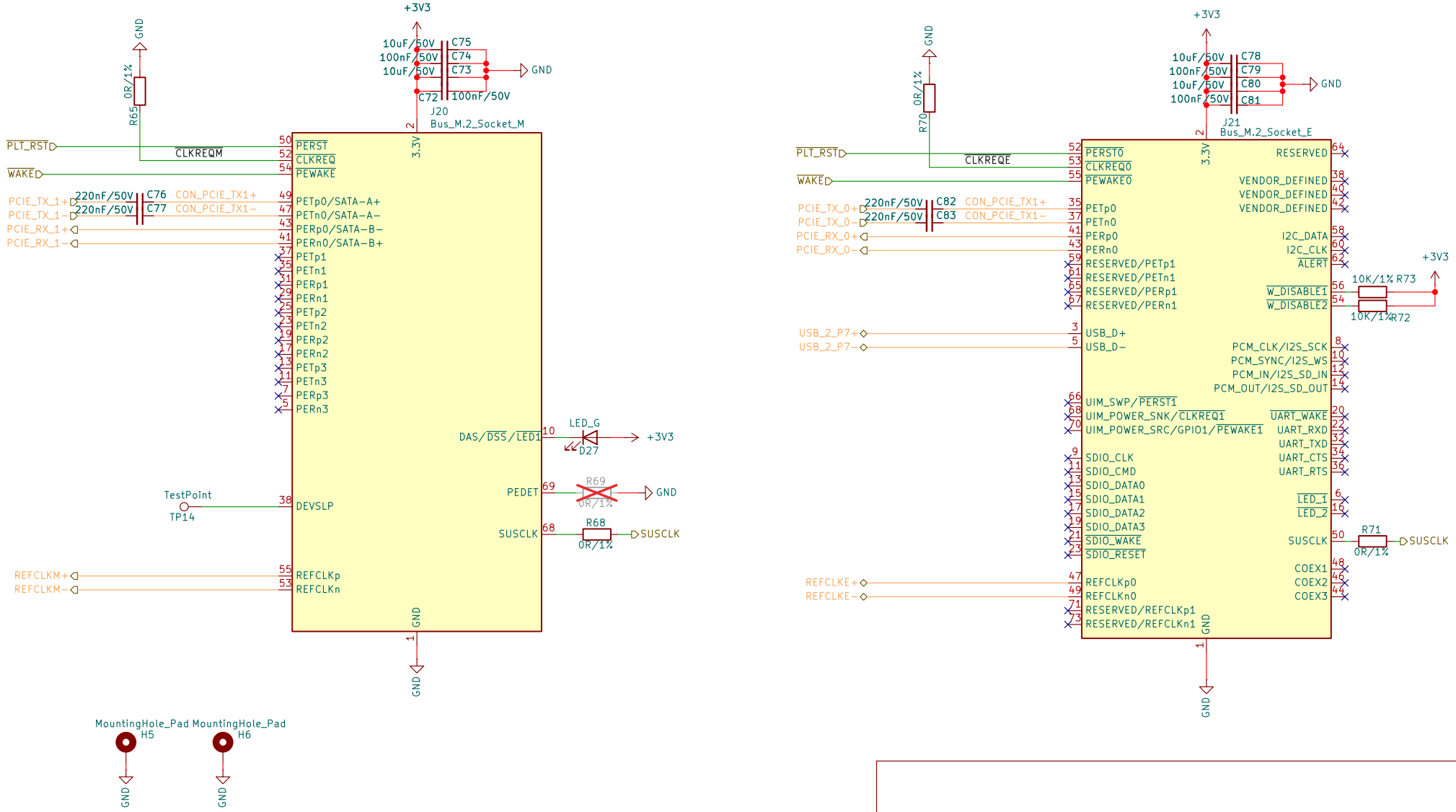


Sheet: /USB 2&3/
File: USB2_3.kicad_sch

Title: practice design

Size: A4	Date: 2026-09-30	**Rev: 1**
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U16
ULC0524P
NC
10
1
2
NC
9
7
4
5
NC
6
TMDS_D2+
TMDS_D2−
TMDS_D1+
TMDS_D1−
VSS
3
GND
U17
ULC0524P
TMDS_D0+
TMDS_D0−
TMDS_CK+
TMDS_CK−
DDI_B_TX0+
DDI_B_TX0−
DDI_B_TX1+
DDI_B_TX1−
DDI_B_TX2+
DDI_B_TX2−
DDI_B_TX3+
DDI_B_TX3−
100nF/50V
C63
C64
C65
C66
C67
C68
C69
C70
470R/1%
R51
R52
R53
R54
R55
R56
R57
R58
5V_HDMI_VCC
C71
F5
6V/50mA
+5V
18
J19
HDMI_A
D2+
D2−
D1+
D1−
D0+
D0−
CK+
CK−
CEC
SCL
SDA
UTILITY
HPD
D2S
D1S
D0S
CKS
SH
PWR_FLAG
D23
1N5819WS
Q9
2SK3018
R59
2.5K/1%
PWR_LED/PSON
+3V3
R60
R61
R62
R63
2.2K/1%
Q6
Q7
DDI_B_DDC_SCL
DDI_B_DDC_SDA
DDC_SCL
DDC_SDA
D25
D24
ESD5V0B03
Q8
2N7002
DDI_B_HPD
R64
1M/1%
R66
1K/1%
HPD
R67
21.5K/1%
D26
Sheet: /HDMI/
File: HDMI.kicad_sch
Title: practice design
Size: A4
Date: 2026-09-30
KiCad E.D.A. 10.0.1
Rev: 1
Id: 9/10



Sheet: /M2 M Key/		
File: M2MKey.kicad_sch		
Title: practice design		
Size: A4	Date: 2026-09-30	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 10/10